Nonlinear Photoacoustic Imaging Using Femtosecond Burst Excitation

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Abstract: We propose a method for multiphoton photoacoustic imaging using GHz-burst-mode femtosecond lasers. Conventional ultrafast systems cannot deliver sufficient energy for photoacoustic signal generation despite achieving high peak intensities. By packaging hundreds of femtosecond pulses within the stress confinement time, burst-mode operation accumulates energy while maintaining nonlinear excitation conditions. We derive generalized enhancement factors for n-photon absorption and free-carrier-assisted regimes, predicting enhancements of 10^5–10^9 over nanosecond excitation. Numerical simulations using silicon nanoparticles as a nonlinear contrast agent in a scattering tissue model validate these predictions, demonstrating the feasibility of deep-tissue nonlinear photoacoustic imaging.

1. Introduction

Deep optical resolution plays a crucial role in biological sciences, and many imaging modalities have been developed to achieve high lateral resolution at deeper sections of biological tissue [1]. High resolution microscopy for sub mm range is clinically significant for the determination of staging of melanoma tumors, monitoring drug delivery and imaging of tumor cores where drug resistance develops [2–5]. However, there is a fundamental trade-off between high resolution and imaging depth due to optical scattering in complex media [1]. Modalities such as confocal microscopy, stimulated emission depletion (STED), photoactivated localization microscopy (PALM), and super-resolution optical fluctuation imaging (SOFI) can achieve subcellular lateral resolution (< 1 μm), but their imaging depth is limited to approximately 100–200 μm [1]. Optical coherence tomography (OCT), optical coherence microscopy (OCM), and multiphoton microscopy (MPM) can reach depths on the order of 1 mm, but their lateral resolution degrades with depth [1, 6]. Near-infrared (NIR) light scatters less in biological tissue compared to visible light; thus, it can penetrate deeper without undergoing scattering events [7].

Despite NIR light penetrating deeper into tissue, multiphoton microscopy, specifically two-photon microscopy, cannot image beyond 1 mm due to background fluorescence [6]. Upon two-photon (or three-photon) absorption, the emitted fluorescence photons have twice the frequency of the incident photons, placing them in the visible range [8]. During collection, visible photons undergo significantly higher scattering [9]. Thus, even though NIR light can reach deeper sections of biological tissue, collection of the fluorescence photons remains a limiting factor for MPM [9]. Theer demonstrated that the two-photon microscopy depth limitation arises from background fluorescence, which is caused by scattered photons outside the focused region along the beam path [10]. While using longer excitation wavelengths can improve penetration depth, fluorescence-based detection remains inherently limited by collection efficiency—motivating alternative detection mechanisms [9].

The fluorescence collection problem can be addressed by employing photoacoustic detection [11]. Unlike fluorescence, the generated acoustic waves propagate with minimal scattering, enabling signal detection from deeper regions [12]. Multiphoton absorption can provide optical contrast independent of linear absorption properties, with signal scaling nonlinearly with intensity and providing inherent localization to the focal volume [6]. To resolve this problem, various contrast agents can be employed to increase contrast in the deeper section of the tissue, specifically tuned to have multiphoton absorption at NIR. Notably, exogenous contrast agents are widely used

for clinical applications such as cancer imaging [13], and the NIR window allows usage of many bio-compatible organic and inorganic compounds, offering tunability for certain applications [14]. As an example, silicon nanoparticles are widely used in medicine and biology as a contrast agent for theranostic and diagnostic purposes, such as imaging and therapy for cancer, biosensing, and cell imaging [15–18]. Moreover, their semiconductor nature allows creation of free carriers through nonlinear excitation, where it increases absorption capability of the particle [19, 20].

Multiphoton photoacoustic imaging is highly desirable due to its high optical resolution combined with long-range ultrasound detection capability [21]. Several attempts have been made in the last decade to realize multiphoton photoacoustics; however, these attempts have been largely unsuccessful. Previous studies demonstrated that single femtosecond pulses deliver a low amount of energy compared to conventional nanosecond systems. Even though femtosecond pulses can achieve high optical intensity, they are incapable of producing sufficient acoustic pressure due to the low pulse energy. Since the photoacoustic signal is directly proportional to the absorbed energy density, the multiphoton-induced signal remains too weak for practical imaging [11]. While maintaining pulse intensity below the ablation threshold and total power within thermal damage limits, conventional ultrafast lasers cannot deliver sufficient energy for photoacoustic signal generation [22].

A method combining high energy delivery with nonlinear excitation could enable practical multiphoton photoacoustic imaging. In this work, we propose a method that utilizes an ultrafast GHz-burst-mode laser to generate a strong photoacoustic signal induced by multiphoton absorption [23]. Unlike conventional femtosecond systems limited by single-pulse energy, burst mode accumulates energy from multiple pulses while each individual pulse remains below the ablation threshold, enabling high total energy delivery without sacrificing the high peak intensity required for nonlinear absorption. GHz-burst mode delivers hundreds of femtosecond pulses within a nanosecond time window, ensuring that the total energy delivery time falls within the stress confinement time, τ_{st}, while also satisfying thermal confinement. All pulses contribute to heating of the target tissue, which in turn generates the photoacoustic wave. The resulting photoacoustic pressure P is proportional to the heating function H ($P \propto H$) [24]. Since the photoacoustic wave amplitude is directly related to the amount of heating, the signal can be increased by adding more pulses within the τ_{st} time window. When combined with silicon nanoparticles, further amplification of the resulting photoacoustic signal with consecutive pulsing can be enabled by burst-mode operation. This signal enhancement could enable practical multiphoton photoacoustic imaging.

2. Theory

2.1. Photoacoustic generation

Photoacoustic generation occurs when a modulated optical radiation is absorbed by a body. High-intensity modulated light causes a rapid heat rise, which results in subsequent rapid expansion. This rapid expansion results in the generation of a mechanical stress wave, which is conventionally called a photoacoustic wave. Two primary time scales govern this phenomenon. Thermal confinement time, $\tau_{\mathrm{th}} = d_c^2/4\kappa_{\mathrm{th}}$, is the time scale of the built-up energy dissipated via thermal diffusion. To generate photoacoustic waves, energy needs to be stored before it diffuses thermally. Stress confinement time, $\tau_{\mathrm{st}} = d_c/v_s$, is the time scale of mechanical stress relaxation. Here, d_c is defined as the characteristic dimension and represents the dimensions of the heated region, v_s is the speed of sound, and κ_{th} is the thermal diffusivity in the medium [25].

Photoacoustic generation is governed by a series of differential equations that represent distinct physical mechanisms. The conversion efficiency from optical energy to acoustic pressure is fundamentally governed by the interplay between energy deposition, heat generation, thermal diffusion, and mechanical stress propagation. When both thermal confinement ($\tau \ll \tau_{\mathrm{th}}$) and stress confinement ($\tau \ll \tau_{\mathrm{st}}$) conditions are met, the generalised photoacoustic wave equation is

defined as follows [24]:

$$\nabla^2 p(r,t) - \frac{1}{c^2}\frac{\partial^2 p(r,t)}{\partial t^2} = -\frac{\alpha}{C_p}\frac{\partial H(r,t)}{\partial t} \tag{1}$$

where α is the isobaric volumetric thermal expansion coefficient, C_p is the specific heat capacity at constant pressure, and $H(r,t)$ is the heating function representing the thermal energy deposited per unit volume per unit time. The right-hand side of the equation acts as a source term. The initial rise in the pressure induced by optical excitation when stress and thermal confinement conditions are met can be calculated by the following formula [24]:

$$p_0 = \Gamma \cdot H \tag{2}$$

H is the volumetric heating and Γ is the Grüneisen parameter, which reflects the photoacoustic characteristic of the target, defined as $\Gamma = \alpha v_s^2/C_p$, where α is the isobaric volume expansion coefficient in K^{-1}, C_p is the specific heat capacity in J/(K·kg). The heating function H, under purely linear optical absorption, is given by $H = \mu_a F$, where μ_a is the linear absorption coefficient (cm^{-1}) and F is the fluence of the laser beam that delivers optical energy.

Modern ultrafast lasers can produce intensities up to $\mathrm{TW/cm}^2$ [26]. At these elevated intensity levels the linear absorption model breaks down, and the material's response becomes nonlinear [22]. To include nonlinear effects, the heating function needs further modification. We employ intensity since the multiphoton excitation rate scales with the intensity of the applied electromagnetic field. Assuming the incident laser beam's spatial form as Gaussian, one can approximate the intensity of the field as $F \approx I \cdot \tau$ where I is the average intensity during the laser pulse of temporal length τ. For two-photon absorption (TPA), which is the dominant nonlinear process for the materials and wavelengths considered in this work, the heating function becomes:

$$H = \left(\mu_a I + \beta I^2\right)\tau \tag{3}$$

where β is the two-photon absorption coefficient with units of cm/W. At sufficiently high intensities reaching gigawatts per centimeter square, the nonlinear term begins to dominate depending on the material properties. Note that this formulation also includes a conventional linear absorption term and is also valid at lower intensity levels. Inserting the heating function into the initial pressure relation of Eq. (2) yields:

$$p_0 = \Gamma\left(\mu_a I + \beta I^2\right)\tau \tag{4}$$

2.2. Free carrier effects

The assumption of a constant absorption coefficient in Eq. (3) needs further modification when a strong laser field induced by an ultrafast burst-mode laser is present, provided the material shows dielectric properties. Early ultrafast pulses within the burst can generate free carriers by excitation of the electrons through the conduction band with multiphoton absorption.

Free carriers are generated via two-photon absorption as follows, assuming $2\hbar\omega > E_g$ where E_g is the band-gap energy of the semiconductor:

$$\frac{dN_{\mathrm{fc}}}{dt} = \frac{\beta I^2}{2\hbar\omega} - \frac{N_{\mathrm{fc}}}{\tau_c} \tag{5}$$

where N_{fc} is the free-carrier density, β is the two-photon absorption coefficient, and τ_c is the carrier lifetime [27]. The first term describes carrier generation through nonlinear absorption,

while the second term accounts for recombination. We therefore require $\tau_r \ll \tau_c$ to neglect carrier recombination, and carrier density accumulates over the pulses:

$$N_k \approx \sum_{j=1}^{k} \frac{\beta}{2\hbar\omega} I_j^2 \tau_p \tag{6}$$

where k is the pulse number within the burst and τ_p is the pulse duration. The generated carriers introduce an additional linear absorption channel. The effective absorption coefficient, therefore, becomes [20]:

$$\mu_a^{\text{eff}} = \mu_a + \sigma_{\text{fc}} N_{\text{fc}} \tag{7}$$

where σ_{fc} is the free-carrier absorption (FCA) cross section.

2.3. Burst-mode enhancement

Generating a photoacoustic signal via nonlinear excitation requires high optical intensity levels. While decreasing the pulse duration is the most direct method to increase light intensity, this consequently reduces the energy per pulse, preventing the generation of desired photoacoustic signal levels. To overcome this trade-off, we propose employing burst-mode, high-repetition-rate ultrafast lasers. This approach packages multiple high-intensity ultrafast pulses into a single envelope. By ensuring the total burst duration (τ_{burst}) remains within the stress confinement time ($\tau_{\text{burst}} \ll \tau_{\text{st}}$), a high total energy can be delivered, leading to the generation of high-amplitude photoacoustic pressure.

To compare these scenarios fairly, both must operate within safe intensity limits for biological tissue. The critical intensity for tissue damage scales with pulse duration as $I_c \propto \tau_p^{-\omega}$, where the exponent ω depends on the dominant damage mechanism [28, 29]. For the nanosecond pulse regime, thermal diffusion limits damage ($\omega \approx 0.5$), while in the ultrafast regime, multiphoton ionization dominates and ω becomes $\omega \approx 0.7$–0.8 [30, 31]. Assuming both scenarios operate at equivalent safety margins:

$$I_p = I_0 \left(\frac{\tau_0}{\tau_p}\right)^{\omega} \tag{8}$$

For further calculations, we assume $\omega = 0.75$ [30].

To quantify the enhancement provided by burst-mode operation, we define the enhancement factor $\varepsilon = H_{\text{burst}}/H_{\text{ns}}$. We first derive the general case including FCA, then present the limiting case without carrier effects. Starting with Eq. (5), the carrier density after the k-th pulse follows:

$$N_k = N_{k-1}\, e^{-\tau_r/\tau_c} + \frac{\beta I_p^2 \tau_p}{2\hbar\omega} \tag{9}$$

Assuming $\tau_r \ll \tau_c$ and neglecting recombination, carrier density accumulates as $N_{k-1} \approx (k-1)\,\beta I_p^2 \tau_p / 2\hbar\omega$. The heating from the k-th pulse includes both FCA and direct TPA:

$$H_k = \left[\sigma_{\text{fc}} N_{k-1}\, I_p + \beta I_p^2\right] \tau_p \tag{10}$$

Summing over a burst of N pulses gives:

$$H_{\text{burst}} = \sum_{k=1}^{N} H_k = \frac{N(N-1)}{2}\, \frac{\sigma_{\text{fc}} \beta I_p^3 \tau_p^2}{2\hbar\omega} + N \beta I_p^2 \tau_p \tag{11}$$

where the first term is the cumulative FCA contribution and the second is direct TPA. Comparing with the nanosecond reference $H_{\text{ns}} = \beta I_0^2 \tau_0$ using the scaling law from Eq. (8) and substituting $\omega = 0.75$:

$$\varepsilon \sim N \left(\frac{\tau_0}{\tau_p}\right)^{0.5} + C\, N(N-1) \left(\frac{\tau_0}{\tau_p}\right)^{0.25} \tag{12}$$

where $C = \sigma_{\mathrm{fc}} I_0 \tau_0 / 4\hbar\omega$. The first term is the direct TPA enhancement and the second is the FCA contribution. We can define a critical pulse number:

$$N_c \sim \frac{4\hbar\omega}{\sigma_{\mathrm{fc}} I_p \tau_p} \tag{13}$$

For $N > N_c$, the FCA term dominates over direct TPA.

When FCA is negligible (e.g., in non-semiconductor targets or for small burst counts $N \ll N_c$), the enhancement reduces to:

$$\varepsilon \approx N \sqrt{\frac{\tau_0}{\tau_p}} \tag{14}$$

3. Simulation study

To validate the theoretical enhancement predictions derived in Section 2 and demonstrate their practical applicability, a simulation study was conducted using realistic biological imaging parameters. To test this hypothesis under controlled conditions, we employed silicon nanoparticles (SiNP) as a nonlinear contrast agent. For the acoustic propagation simulations, we use the k-Wave toolbox for MATLAB, which solves the acoustic wave equations using a k-space pseudo-spectral method, enabling efficient and accurate simulation of photoacoustic wave propagation in homogeneous media [32].

Table 1. System parameters used in analytical calculations.

Parameter	Symbol	Value	Unit
Wavelength	λ	1064	nm
Numerical aperture	NA	0.50	–
Refractive index	n	1.33	–
FS pulse duration	τ_p	100	fs
FS focal fluence	F_p	0.1	J/cm^2
Burst window	τ_{burst}	1	ns
NS pulse duration	τ_0	1	ns
NS focal fluence	F_0	1.0	J/cm^2
Absorption coeff. (tissue)	μ_a	0.05	mm^{-1}
Scattering coeff. (tissue)	μ_s	5.6	mm^{-1}
Grüneisen parameter	Γ	0.12	–
TPA coefficient (c-Si)	β	1.5×10^{-11}	m/W
FCA cross section	σ_{fc}	5×10^{-22}	m^2

Tissue: native grey matter at 1100 nm [33]; c-Si properties at 1064 nm [20].

3.1. Medium and contrast agent

To establish a realistic medium configuration, the surrounding medium properties were chosen as biological tissue, with optical properties taken from experimentally reported native brain grey matter absorption and scattering data at 1100 nm [33]. The absorption coefficient of native

brain grey matter is $\mu_a = 0.05\,\mathrm{mm}^{-1}$, and scattering $\mu_s = 5.6\,\mathrm{mm}^{-1}$. The mean free path l_t is defined as $l_t = 1/\mu_s$, where $g = 0.92$ is the anisotropy coefficient, giving $l_t \approx 0.18$ mm for our configuration. We use $\mu_t = \mu_a + \mu_s$ as the attenuation coefficient since we count ballistic photons reaching the target.

The target is selected as silicon nanoparticles (SiNP), which are a common contrast agent in nonlinear imaging applications with well-defined optical properties and the capability to produce free carriers [19, 20]. SiNPs have $\beta = 1.5$ cm/GW and $\sigma_{\mathrm{fca}} = 5 \times 10^{-18}$ cm^2 [20].

3.2. Optical model

The initial pressure distribution $p_0(\mathbf{r})$ was calculated by modeling the optical excitation and energy deposition within the heterogeneous medium. Our model accounts for realistic beam propagation effects, including absorption and scattering. The excitation laser beam was modeled as a Gaussian profile spatially and focused to the target at varying depths starting from 100 µm to 3 mm with 500 µm increments. In nonlinear deep tissue imaging applications, the sweet spot for NA is 0.6–0.8, which creates a tighter spot size and achieves better lateral and axial resolution [34]. We modeled focusing with a lens having a long working distance and a high numerical aperture NA = 0.50, with $\omega_0 = 0.68$ µm for $\lambda = 1064$ nm. Light propagation through the medium was simulated using a layer-by-layer approach, where at each depth z, the Gaussian beam waist $\omega(z)$ was calculated using standard Gaussian formalism:

$$w(z) = w_0 \sqrt{1 + \left(\frac{z - z_{\mathrm{focus}}}{z_R}\right)^2} \tag{15}$$

This approach allows focusing at depth z_{target} where $\omega(z_{\mathrm{target}}) = \omega_0$ ensures tight focusing at the target location. Input peak surface intensity was calculated with input fluence and pulse duration, with the approximation $I \approx F \cdot \tau$. Using the calculated beam waist $\omega(z)$, the intensity map at each depth is:

$$I(y, z) = I_{\mathrm{peak}}(z) \cdot e^{-2y^2/w(z)^2} \tag{16}$$

Including Beer–Lambert attenuation through propagation:

$$\frac{dI}{dz} = -\mu_t(x, y) \cdot I \tag{17}$$

We used the linear part of the Beer–Lambert law with Taylor series expansion for sufficiently small step size Δz:

$$I_{\mathrm{peak}}(z + \Delta z) = I_{\mathrm{peak}}(z) \cdot (1 - \mu_t \Delta z) \tag{18}$$

where μ_t is the attenuation coefficient for the surrounding medium. Attenuation due to nonlinear absorption is negligible because the surrounding medium is water-dominated, and water has low linear absorption near 532 nm, which corresponds to two-photon absorption at 1064 nm [35]. Moreover, since the beam is tightly focused on the target, the Rayleigh range $z_R = 1.8$ µm is short, so we can ignore the I^2 term in the attenuation. After intensity is mapped at each location, the generated heat at each cell can be calculated via Eq. (11) for burst scenarios and Eq. (20) for the single nanosecond pulse.

The initial pressure distribution was then obtained via the Grüneisen relation, valid under conditions of thermal and stress confinement:

$$p_0(x, y) = \Gamma \cdot H(x, y) \tag{19}$$

where $\Gamma = 0.12$ is the Grüneisen parameter for soft tissue.

3.3. Excitation scenarios

Twenty-five excitation scenarios were simulated to demonstrate enhancement through the GHz-PAM modality and compare different cases. To make Eq. (4) valid, the burst duration was set to $\tau_b = \tau_{\text{ns}} = 1$ ns; however, $\tau_{\text{stress}} \approx 0.9$ ns for $\omega_0 = 0.68$ µm spot size and the nanosecond pulse is barely outside the stress confinement region. Individual femtosecond pulses have $\tau_{\text{fs}} = 100$ fs, which satisfies the stress confinement condition. To calculate the delivered heat by the burst, we assume $Q_{\text{burst}} = N \cdot Q_{\text{individual}}$, valid since all pulses are delivered within the stress confinement time. The intra-burst repetition rate was varied by keeping τ_b constant at 100, 500, and 1000 GHz, corresponding to $N = 100, 500, 1000$ pulses within the burst duration.

Fluence for the single femtosecond pulse, $F_p = 0.1$ J/cm^2, was set to deliver 5% of the maximum intensity without passing the ablation threshold for human tissue [30]. Using Eq. (8), fluence for a single nanosecond pulse is calculated approximately as $F_0 = 1$ J/cm^2. For given pulse durations, femtosecond pulses have energy $E_{p,\text{fs}} = 0.72$ nJ and the nanosecond pulse has $E_{p,\text{ns}} = 7.2$ nJ. Similarly, the femtosecond pulse reaches the focal intensity $I_p = 1$ TW/cm^2 while the nanosecond pulse produces $I_0 = 1$ GW/cm^2.

4. Results

Results of the simulations are presented in Figs. 1–5.

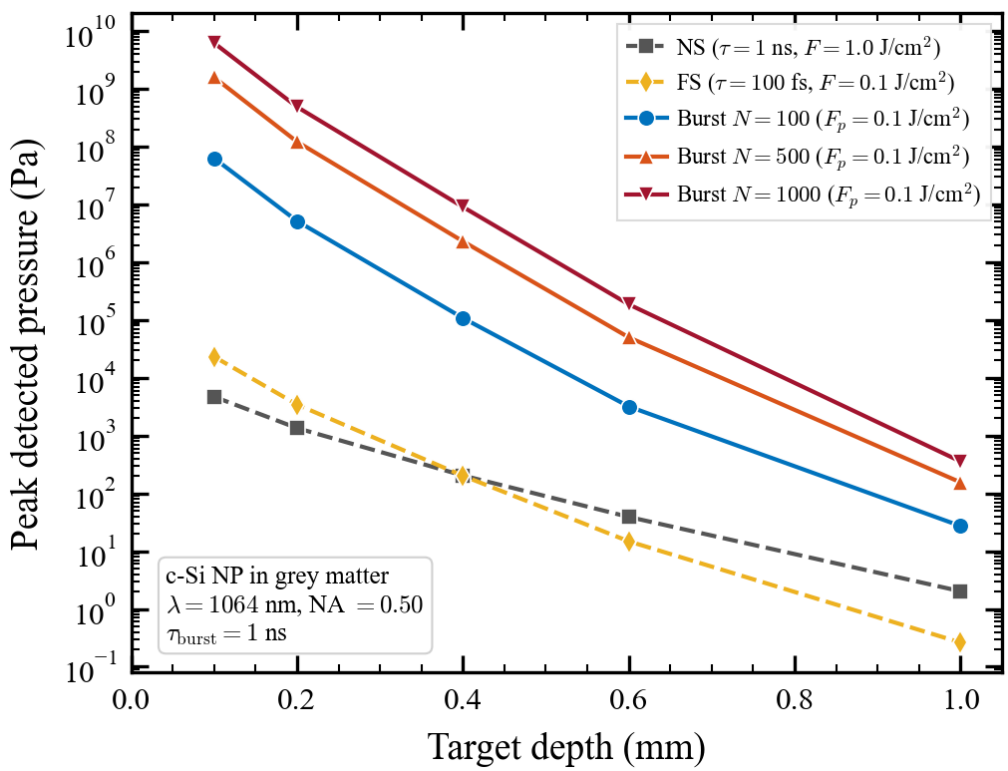

Fig. 1. Peak detected photoacoustic pressure as a function of target depth for five excitation scenarios.

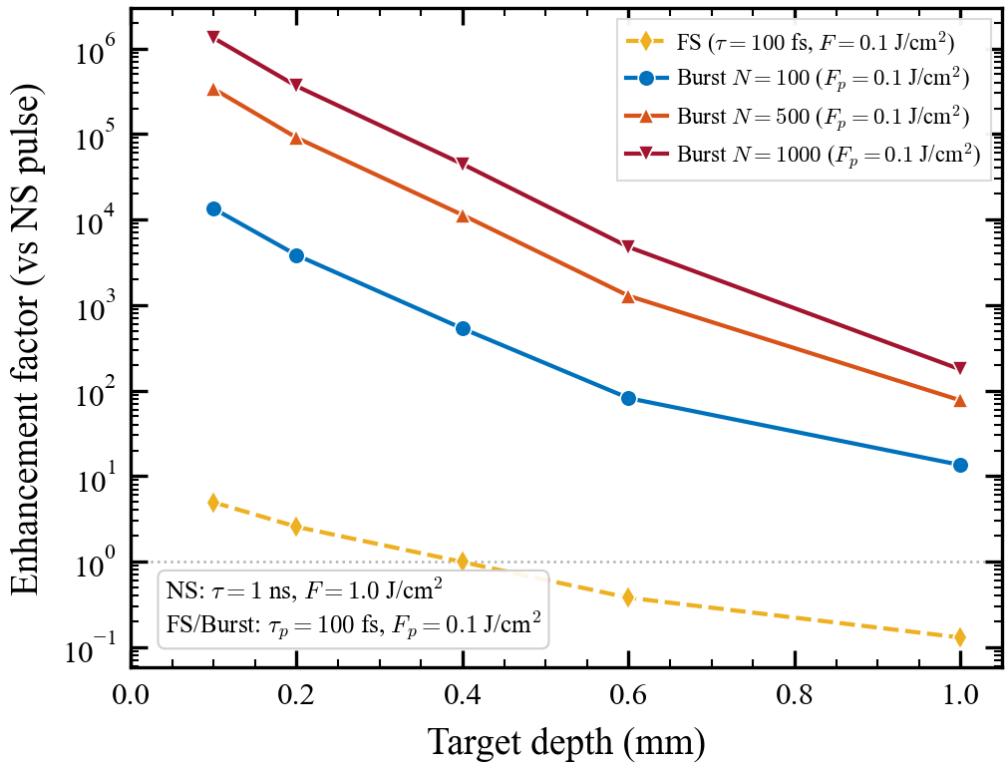

Fig. 2. Enhancement factor of each excitation scenario relative to the nanosecond reference pulse as a function of depth.

5. Discussion

To create high intensities to induce multiphoton absorption effects, narrow spot-size configurations are necessary. Reducing the spot size results in a smaller light–material interaction surface, which determines d_c. As demonstrated in the simulation, the stress confinement condition is challenging to achieve for nonlinear photoacoustic imaging applications.

Lower pulse energies increase the feasibility for future imaging applications, allowing higher repetition rates without entering the thermal damage regime, which is determined around 2.5 mW for nonlinear imaging applications [36]. However, at relatively high intensities produced by ultrafast lasers (GW/cm^2), optical breakdown occurs as plasma starts to form due to avalanche ionization [22]. This process is named *plasma-induced ablation* and results in rapid vaporization of the incident surface [37]. Since the proposed application is designed to operate at high intensities close to the ablation-threshold regime, pulse energies need to be set carefully.

Laser-induced optical breakdown exhibits different pulse-scaling behavior for different pulse regimes. In the nanosecond regime where $\tau > 500$ ps, the fluence threshold shows an $F_{\text{th}} \propto \tau^{0.5}$ relationship. In the picosecond regime, 1 ps $< \tau <$ 500 ps, deviations from the $\tau^{0.5}$ curve begin and the curve presents a relationship ω = 0.6–0.8 depending on the material or tissue. The ablation threshold starts to decrease slowly in this regime, as shown experimentally for biological tissues in many studies [28, 30, 31]. After entering the $\tau < 1$ ps regime, the relationship enters a plateau regime where F_{th} remains constant due to negligible thermal and collision losses during the ultrashort interaction time [38].

When both pulses operate near their threshold intensity levels, Eqs. (12) and (14) allow us to

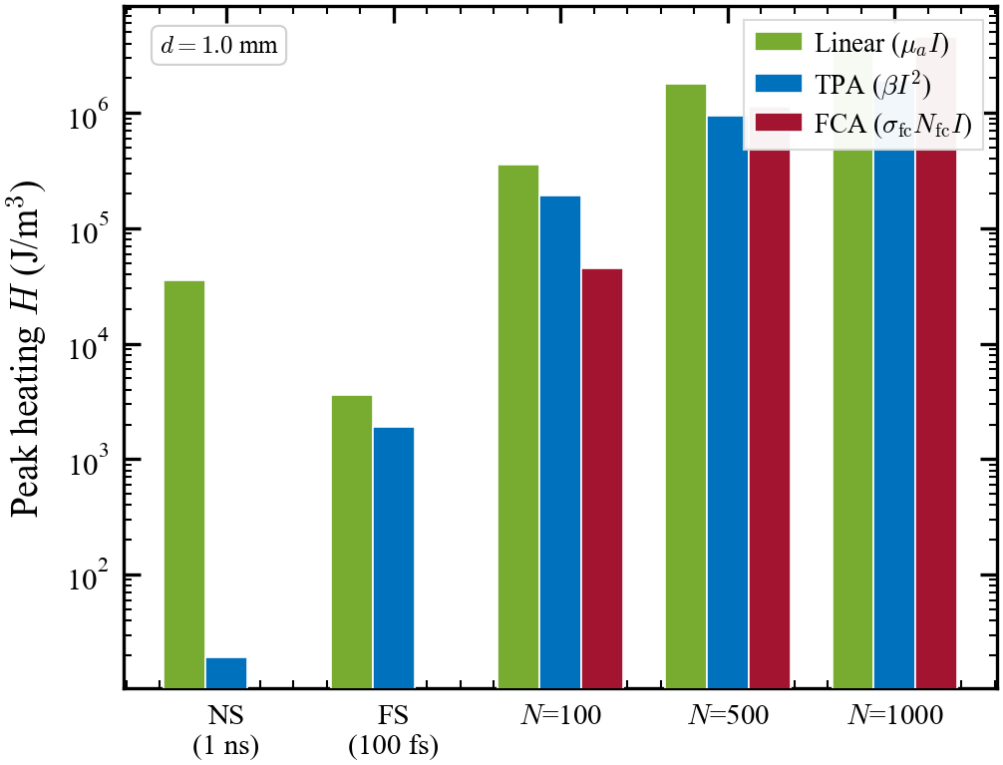

Fig. 3. Comparison of linear, TPA, and FCA contributions to peak heating at the focal volume for each excitation scenario at a fixed depth.

project the enhancement generated by GHz-burst-mode operation. However, operation at near ablation-threshold intensities is non-trivial since photoacoustic imaging applications are targeted for *in vivo* imaging and biological tissue damage is governed by many different mechanisms, including photothermal damage and plasma-induced ablation [37]. Plasma-induced ablation is caused by high-intensity optical fields produced by ultrafast lasers (GW/cm^2). At these intensity levels, optical breakdown occurs as plasma starts to form due to avalanche ionization, resulting in rapid vaporization of the incident surface [22, 37]. Since the individual pulses interact with the tissue shorter than 1 μs, thermal diffusion is assumed to be negligible during individual pulse delivery [37]. However, since pulsed operation delivers many pulses to the interaction surface, heat accumulates, and thermal damage also starts to act as another limiting factor for the amount of energy delivered. For nonlinear imaging applications, the thermal damage threshold is determined around 2.5 mW for safe operation [36].

A. Detailed enhancement derivations

A.1. Two-photon absorption enhancement

Here we present the full algebraic derivation of the TPA enhancement factor stated in Eq. (14). From Eq. (3), the energy deposited per pulse is:

$$H = (\mu_a I + \beta I^2)\tau \tag{20}$$

For a nanosecond pulse with intensity I_0 and duration τ_0, the delivered total energy is $H_{\text{ns}} = (\mu_a I_0 + \beta I_0^2)\tau_0$. For a femtosecond burst with intensity I_p, pulse duration τ_p, and N pulses, the

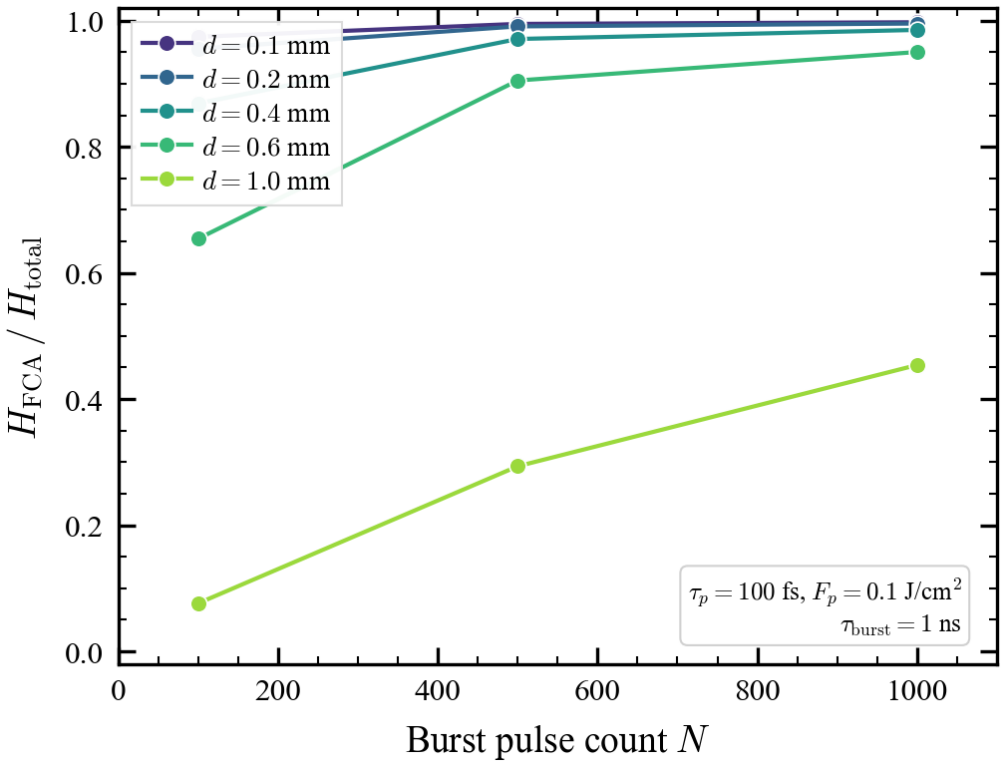

Fig. 4. Fraction of total heating attributed to FCA as a function of burst pulse count for different target depths.

total energy is $H_{\mathrm{burst}} = N(\mu_a I_p + \beta I_p^2)\tau_p$.

The enhancement factor is:

$$\varepsilon = \frac{H_{\mathrm{burst}}}{H_{\mathrm{ns}}} = \frac{N(\mu_a I_p + \beta I_p^2)\tau_p}{(\mu_a I_0 + \beta I_0^2)\tau_0} \tag{21}$$

Substituting Eq. (8) into Eq. (21) yields:

$$\begin{aligned}\varepsilon &= N \times \frac{\mu_a I_0 \left(\frac{\tau_0}{\tau_p}\right)^{\omega} + \beta I_0^2 \left(\frac{\tau_0}{\tau_p}\right)^{2\omega}}{(\mu_a I_0 + \beta I_0^2)} \times \frac{\tau_p}{\tau_0} \\ &= N \times \frac{\mu_a \left(\frac{\tau_0}{\tau_p}\right)^{\omega-1} + \beta I_0 \left(\frac{\tau_0}{\tau_p}\right)^{2\omega-1}}{(\mu_a + \beta I_0)}\end{aligned} \tag{22}$$

When nonlinear absorption dominates ($\beta I_0 \gg \mu_a$), with $\omega = 0.75$, Eq. (22) simplifies to Eq. (14).

A.2. Generalized n-photon enhancement

The two-photon framework presented in the main text can be generalized to n-photon absorption by replacing βI^2 with $\beta_n I^n$, where β_n is the n-photon absorption coefficient. When n-photon absorption dominates:

$$\varepsilon = \frac{N\beta_n I_p^n \tau_p}{\beta_n I_0^n \tau_0} \tag{23}$$

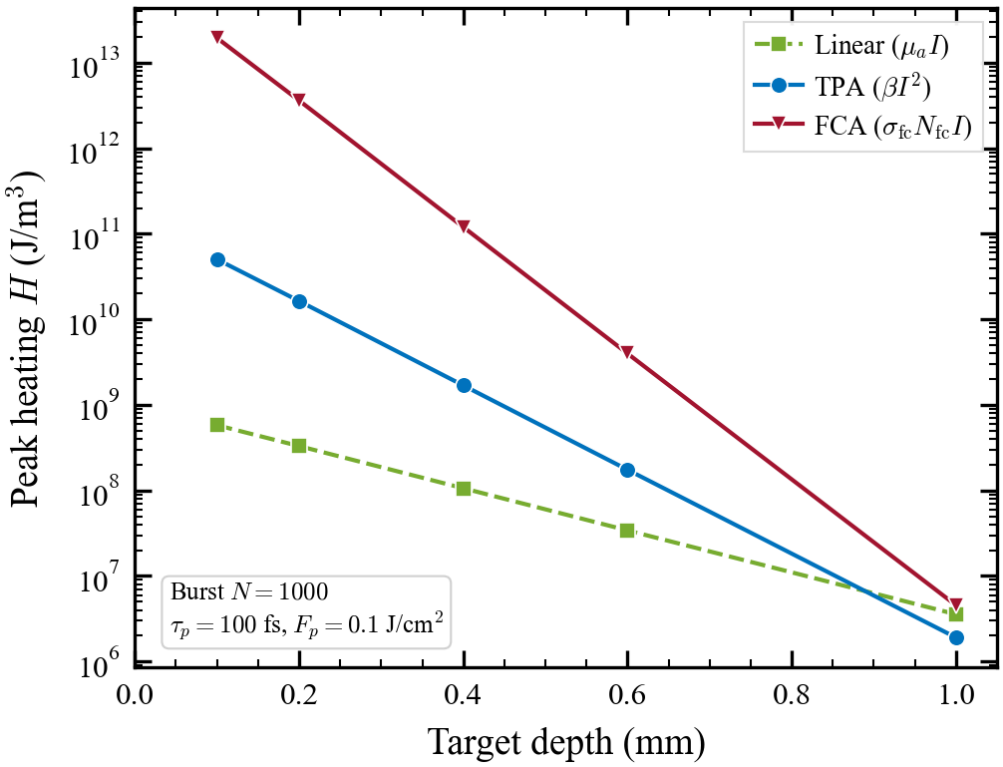

Fig. 5. Depth dependence of peak linear, TPA, and FCA heating contributions for the $N = 1000$ burst configuration.

Using the intensity scaling relation $I_p/I_0 = (\tau_0/\tau_p)^\omega$ and canceling β_n:

$$\varepsilon = N \left(\frac{\tau_0}{\tau_p}\right)^{n\omega-1} \tag{24}$$

For $n = 2$ and $\omega = 0.75$, this recovers Eq. (14). For three-photon absorption ($n = 3$):

$$\varepsilon = N \left(\frac{\tau_0}{\tau_p}\right)^{1.25} \tag{25}$$

Similarly, the FCA enhancement from Eq. (12) generalizes to:

$$\varepsilon \sim N \left(\frac{\tau_0}{\tau_p}\right)^{n\omega-1} + C_n\, N(N-1) \left(\frac{\tau_0}{\tau_p}\right)^{(n+1)\omega-2} \tag{26}$$

where $C_n = \sigma_{\rm fc} I_0 \tau_0 / 2n\hbar\omega$.

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Data Availability Statement. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

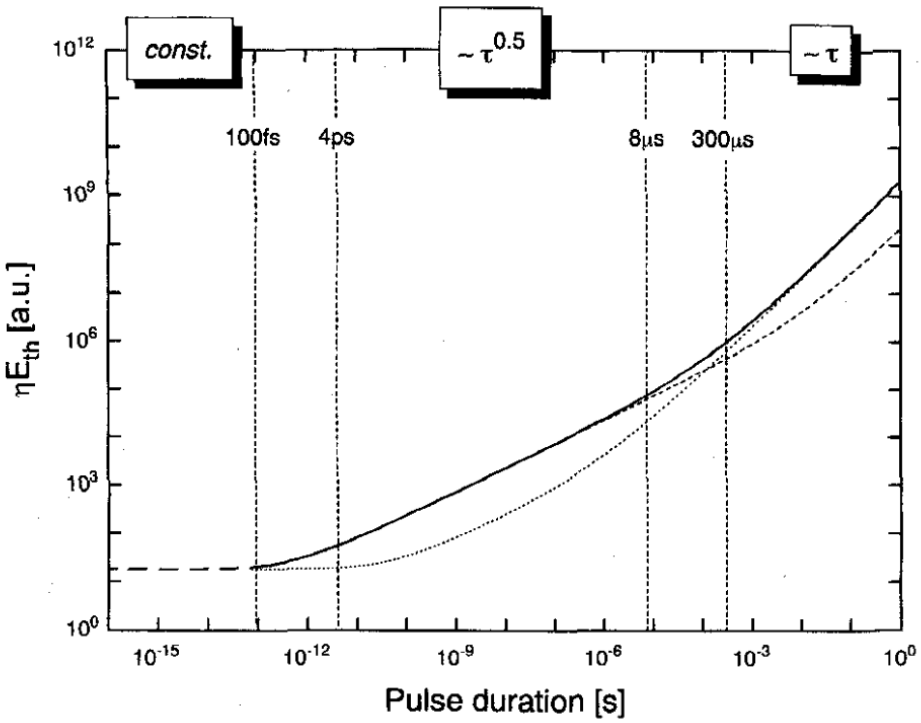

Fig. 6. Pulse duration dependence of breakdown threshold from Loesel *et al.* [38]. The model includes collision (τ_c) and diffusion (τ_d) time constants; the solid line uses $\tau_c = 1$ fs and $\tau_d = 500$ ps, showing the characteristic square-root dependence in the nanosecond-to-picosecond regime.

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